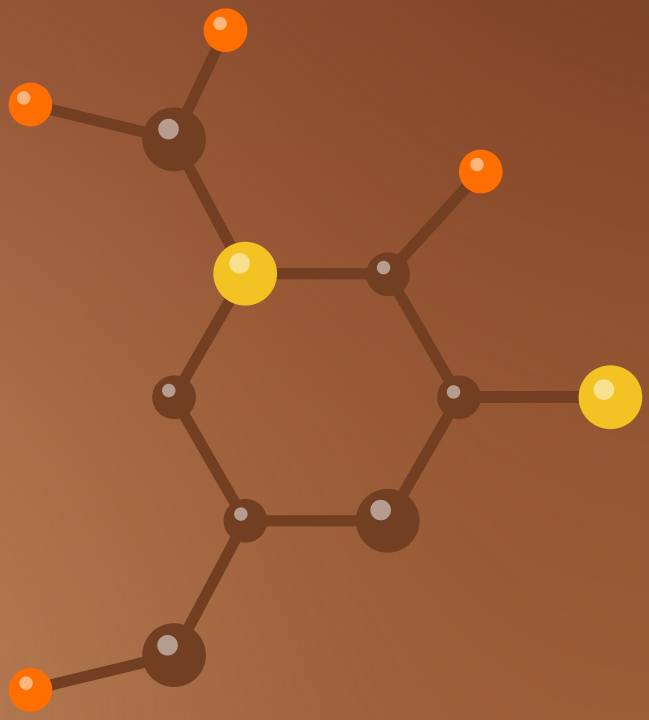
ALLEN CAREER INSTITUTE
path to success
KOTA (RAJASTHAN)



# ORGANIC CHEMISTRY

## JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

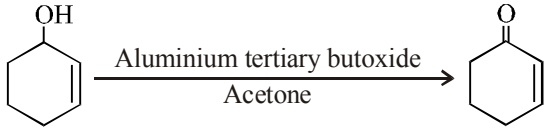
**EXERCISE**

## Oxidation

English Medium

## EXERCISE#O-1

**1.** The given reaction



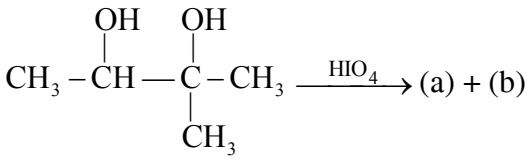

is knows as :

(A) Kolbe reaction (B) Tischenko reaction

(C) MPV reaction (D) Oppeneur oxidation

**OX0001**

**2.** In the given reaction :

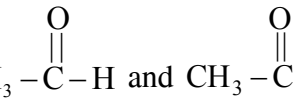
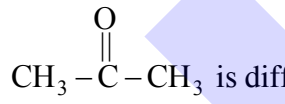



(a) and (b) respectively be :

(A) $CH_3CHO$ and $CH_3CHO$ (B) $CH_3COCH_3$ and $CH_3CHO$

(C) $CH_3COCH_3$ and $CH_3COCH_3$ (D) $CH_3COOH$ and $CH_3COCH_3$

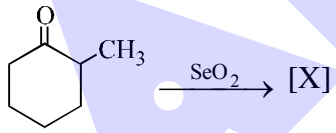
**OX0002**

**3.**  and  is differentiated by

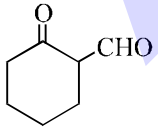
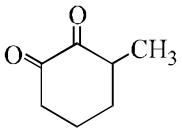
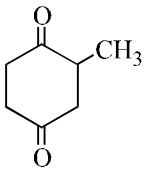
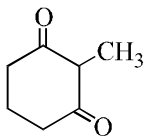
(A) Tollen's reagent (B) Lucas test (C) Iodoform test (D) $NaHSO_3$ test

**OX0003**

**4.** In the given reaction




[X] will be :

(A)  (B)  (C)  (D) 

**OX0004**

**5.** Which will give silver mirror test with Tollens reagent :

(A) $C_6H_5CHO$ (B) $CH_3$–CHO (C) HCOOH (D) All of these

**OX0005**

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**6.** Which one of the following compounds will give dimethyl glyoxal with $SeO_2$:

(A) Acetone (B) Acetophenone

(C) Ethyl methyl ketone (D) Propanaldehyde

**OX0006**

**7.** Which one of the following alcohols can be oxidised by $K_2CrO_4$?

(A) Ethanol (B) Tert butyl alcohol

(C) Isopropyl alcohol (D) Allyl alcohol

**OX0007**

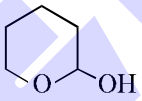
**8.** Fehling solution gives red precipitate with :

(A) Aromatic aldehyde (B) Aliphatic aldehyde

(C) Ketones (D) $CH_2(OH)-C(=O)-CH_3$

**OX0008**

**9.** Which of the following compound will give positive Tollens test

(A) $CH_3CHO$ (B) 

(C) $CH_3CH_2CH_2CH_2CH(OH)-OCH_3$ (D) $CH_3-CH(OCH_3)_2$ **OX0009**

**10.** Schiff's reagent gives pink colour with :

(A) Acetaldehyde (B) Formic acid (C) Acetic acid (D) Methyl acetate

**OX0010**

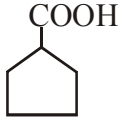
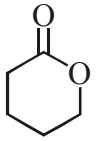
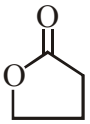
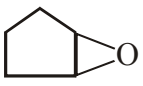
**11.** Which one of the following alcohols are oxidised by $MnO_2$?

(A) $C_6H_5-CH_2-CH_2-OH$ (B) $CH_2=CH-CH_2-CH(OH)-CH_3$

(C) $CH_3-CH=CH-CH(OH)-CH_3$ (D) $CH_3-CH_2-CH_2-OH$

**OX0011**

**12.**  Major product is :

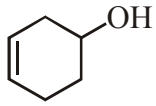
(A)  (B)  (C)  (D) 

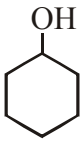
**OX0012**

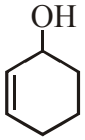


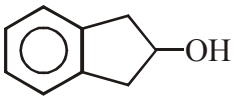
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**13.** Which of the following can be oxidised by $MnO_2$:

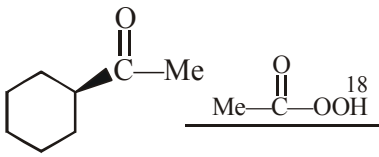
(A) 

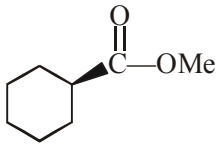
(B) 

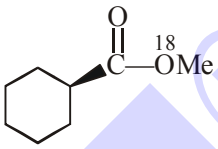
(C) 

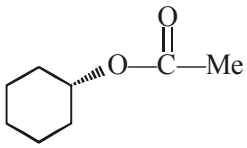
(D) 

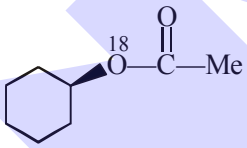
**OX0013**

**14.**  Major product is :

(A) 

(B) 

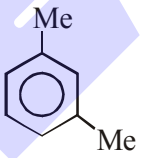
(C) 

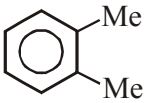
(D) 

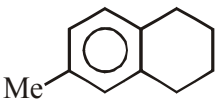
**OX0014**

**15.** A $\xrightarrow{H^+ / KMnO_4}$ B $\xrightarrow{\Delta}$ Phthalic Anhydride ; A is :

(A) 

(B) 

(C) 

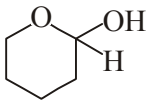
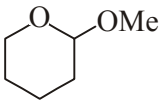
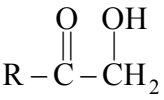
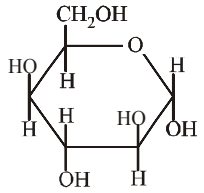
(D) 

**OX0015**

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## EXERCISE#O-2

**1.** Which will give the Tollen test.

(A)  (B)  (C)  (D) 

**OX0016**

**2.** Which of the following is an example of oxidation reaction :

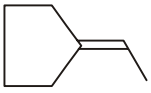
(A) $CH_3 - CH_2 - OH \xrightarrow{PCl_3}$ (B) $CH_3 - CH = CH_2 \xrightarrow{Ni/H_2}$ (C) $CH_3 - CH = CH_2 \xrightarrow[CCl_4]{Br_2}$

**OX0017**

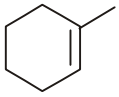
**3.** (i) $CH_2 = CH_2 \xrightarrow{H^\oplus / KMnO_4}$ **OX0018**

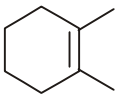
(ii) $CH_3–CH= CH_2 \xrightarrow{H^\oplus / KMnO_4}$ **OX0019**

(iii)  $\xrightarrow{H^\oplus / KMnO_4}$ **OX0020**

(iv)  $\xrightarrow{H^\oplus / KMnO_4}$ **OX0020**

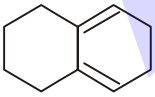
(v)  $\xrightarrow{H^\oplus / KMnO_4}$ **OX0021**

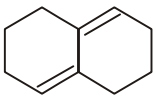
(vi)  $\xrightarrow{H^\oplus / KMnO_4}$ **OX0021**

(vii)  $\xrightarrow{H^\oplus / KMnO_4}$ **OX0022**

(viii)  $\xrightarrow[-CO_2]{H^\oplus / KMnO_4, \Delta}$ **OX0022**

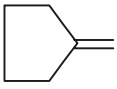
(ix)  $\xrightarrow{H^\oplus / KMnO_4}$ **OX0023**

(x)  $\xrightarrow{H^\oplus / KMnO_4}$ **OX0023**

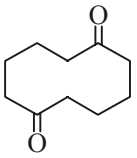
(xi)  $\xrightarrow{H^\oplus / KMnO_4}$ **OX0023**

**4.** A to F alkenes with minimum possible carbon.

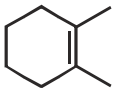
(i) A $\xrightarrow[\Delta]{H^\oplus / KMnO_4}$ MeCOOH as the only product **OX0024**

(ii) B $\xrightarrow[\Delta]{H^\oplus / KMnO_4}$  as the only product **OX0024**

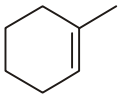
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hdb-Alc & Et- Hydr - Carb -Oxi - Red - Ald & Ket-(OC)\Eng\05_Oxidation\01_Theory+Ex.p65"

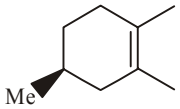
(iii) D $\xrightarrow[\Delta]{H^{\oplus}/KMnO_4}$  **OX0025**

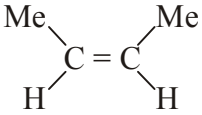
(iv) F $\xrightarrow[\Delta]{H^{\oplus}/KMnO_4}$ Acetone + Ethanoic acid **OX0025**

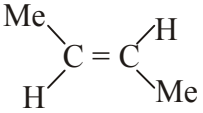
**5.** (i)  $\xrightarrow[KMnO_4]{1\%\text{ alkaline}}$ A

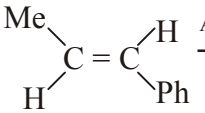
$\xrightarrow{m\text{-CPBA}/H_3O^+}$ B **OX0026**

(ii)  $\xrightarrow{mCPBA}$ **OX0026**

(iii)  $\xrightarrow[\text{hydrolysis}]{mCPBA}$ **OX0027**

(iv)  $\xrightarrow[\text{hydrolysis}]{mCPBA}$ **OX0027**

(v)  $\xrightarrow[\text{hydrolysis}]{mCPBA}$ **OX0028**

(vi)  $\xrightarrow[\Delta]{Ag_2O \text{ or } 2Ag+\frac{1}{2}O_2}$ **OX0028**

**6.** (i) $CH_3-CH_2-CH_2-OH \xrightarrow{KMnO_4/OH^-,\Delta} ?$

$\xrightarrow[\text{or conc. } HNO_3,\Delta]{K_2Cr_2O_7/H^+,\Delta}$

$\text{or } KMnO_4/H^+,\Delta$ **OX0029**

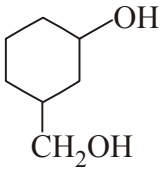
(ii) $CH_3-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_2-CH_3 \xrightarrow[\text{or } K_2Cr_2O_7,H^{\oplus}]{KMnO_4,H^{\oplus}} ?$ **OX0029**

**7.** (i) $CH_2=CH-(CH_2)_3-CH_2-OH \xrightarrow[\text{in glacial acetic acid}]{\text{Chromic anhydride}}$

**OX0030**

(ii) $C_6H_5-CH=CH-CH_2-OH \xrightarrow[\text{in glacial acetic acid}]{\text{Chromic anhydride}}$ **OX0030**

(iii) $CH_3-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_2-CH_2-CH_2-OH \xrightarrow[CH_2Cl_2]{PCC}$ (A) **OX0030**

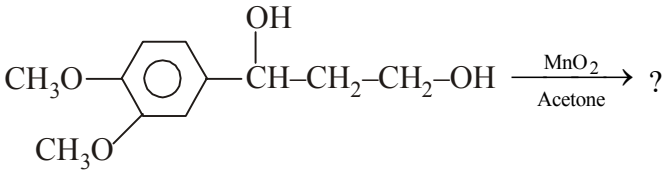
(iv)  $\xrightarrow[CH_2Cl_2]{PCC}$ **OX0031**

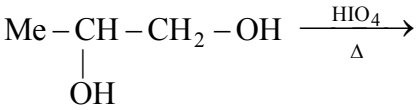
(v) $CH_2=CH-CH_2-OH \xrightarrow{MnO_2} ?$ **OX0031**

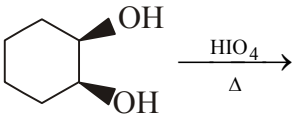
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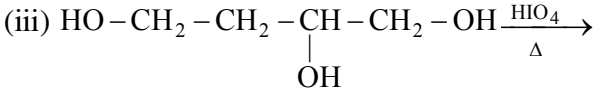


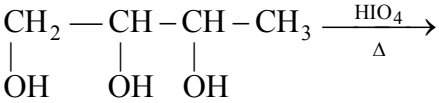
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Halo-Alc & Et - Hydr - Carb - Oxi - Red - Ald & Ket-(OC)\Eng\05_Oxidation\01_Theory+Ex_p65"

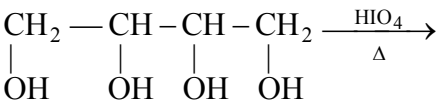
(vi)  **OX0031**

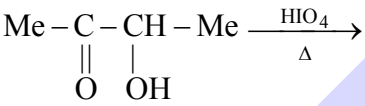
**8.** (i)  **OX0032**

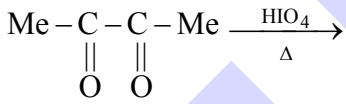
(ii)  **OX0032**

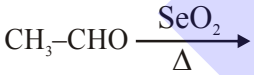
(iii)  **OX0032**

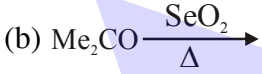
(iv)  **OX0032**

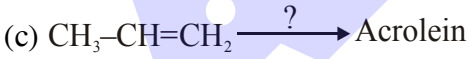
(v)  **OX0033**

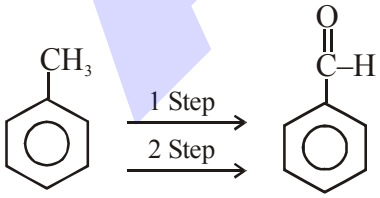
(vi)  **OX0033**

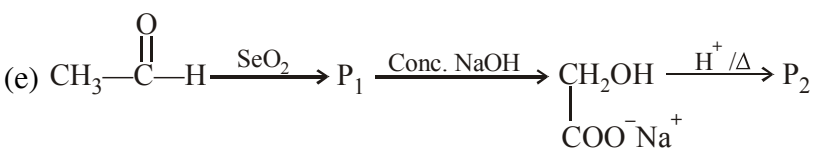
(vii)  **OX0033**

**9.** (a)  **OX0034**

(b)  **OX0034**

(c)  **OX0034**

(d)  **OX0035**

(e)  **OX0036**

**10.** (a) How will you differentiate HCHO and PhCHO ?

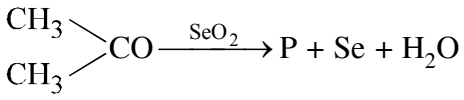
(b) How will you differentiate HCHO and MeCHO ? **OX0037**

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node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hdb-Alc & Et - Hydr - Carb -Oxi - Red - Ald & Ket-(OC)\Eng\05_Oxidation\01_Theory+Ex.p65"

## EXERCISE # MAIN & ADVANCE

**1.** In the reaction, P is **[IIT 1995]**



(A) $CH_3COCHO$ (B) $CH_3COOCH_3$ (C) $CH_3COCH_2OH$ (D) None

**OX0038**

**2.** The best reagent to convert pent–3–en–2–ol into pent –3–en –2–one is - **[AIEEE-2005]**

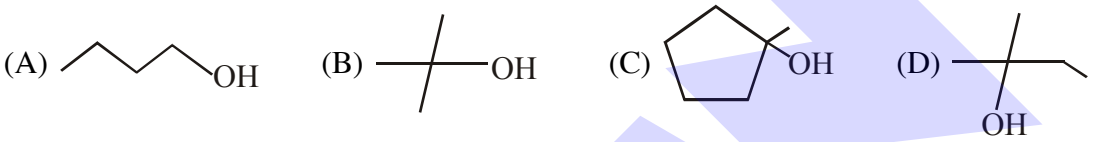
(A) Acidic dichromate (B) Acidic permanganate

(C) Pyridinium chloro-chromate (D) Chromic anhydride in glacial acetic acid

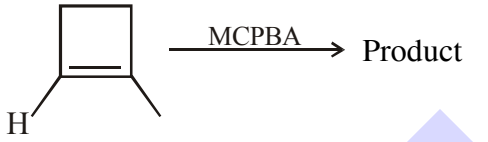
**OX0039**

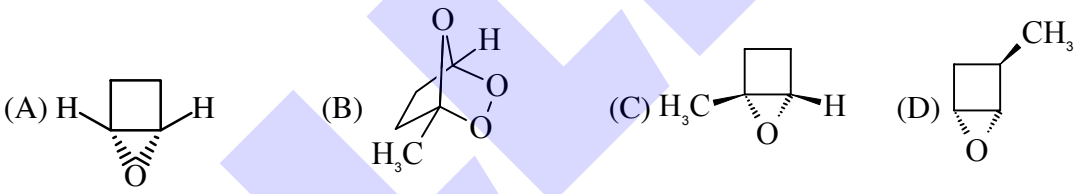
**3.** Which of the following will change the colour of acidic dichromate solution.

**[JEE-MAINS 2013]**

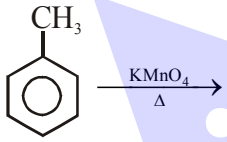


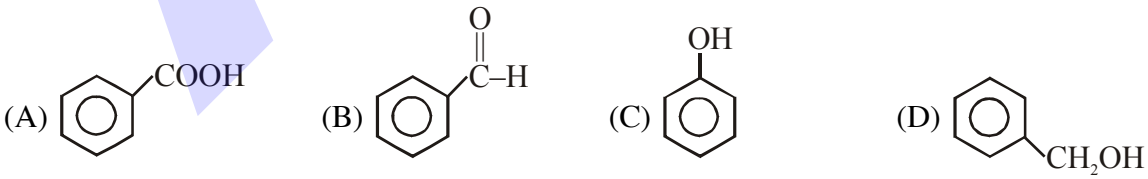
**OX0040**

**4.**  **[JEE-MAINS 2013]**



**OX0041**

**5.**  Major product of reaction is? **[JEE-MAINS 2013]**



**OX0042**

**6.** The most suitable reagent for the conversion of $R–CH_2–OH \rightarrow R–CHO$ is :-

**[JEE-MAINS 2014]**

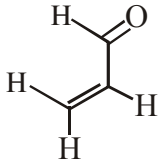
(A) $CrO_3$ (B) PCC (Pyridinium chlorochromate)

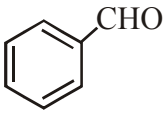
(C) $KMNO_4$ (D) $K_2Cr_2O_7$

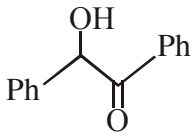
**OX0043**

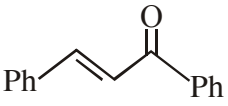
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#\Halo-Alc.&Et.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket-(OC)\Eng\05_Oxidation\01_Theory+Ex_p65

**7.** Positive Tollen's test is observed for **[JEE-Advance 2016]**

(A) 

(B) 

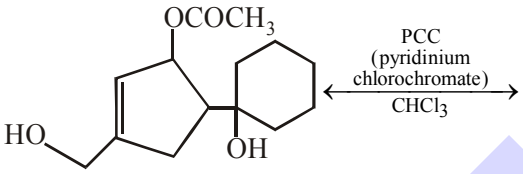
(C) 

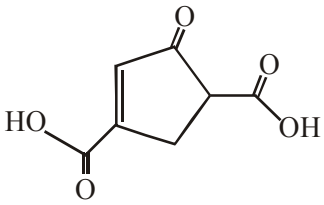
(D) 

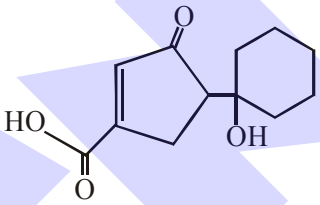
**OX0044**

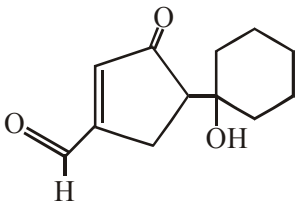
**8.** The major product formed in the following reaction is :-

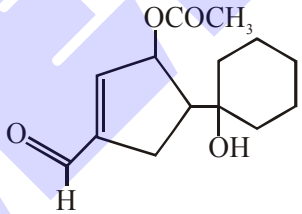
**[Jee Main - On_line Evening 2018]**



(A) 

(B) 

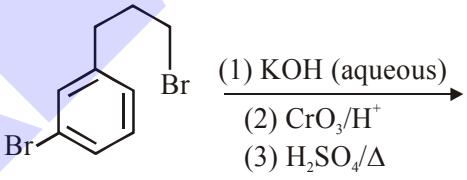
(C) 

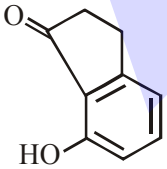
(D) 

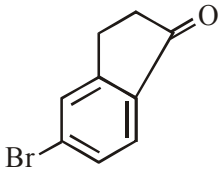
**OX0045**

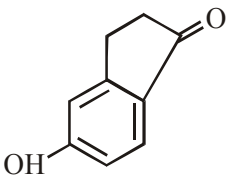
**9.** The major product of the following reaction is:

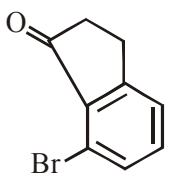
**[Jee-Main-On-line-(Jan)-2019]**



(1) 

(2) 

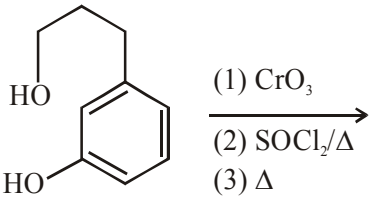
(3) 

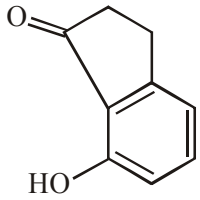
(4) 

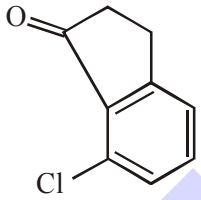
**OX0046**

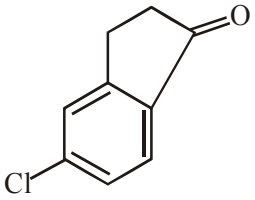
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hdlo-Alc & Et- Hydr - Carb -Oxi - Red - Ald & Ket-(OC)\Eng\05_Oxidation\01_Theory+Ex.p65"

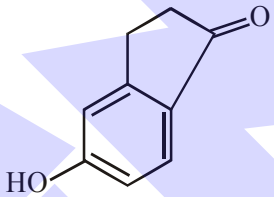
**10.** The major product of the following reaction **[JEE-MAIN-On-line-(April)-2019]**




(1) 


(2) 

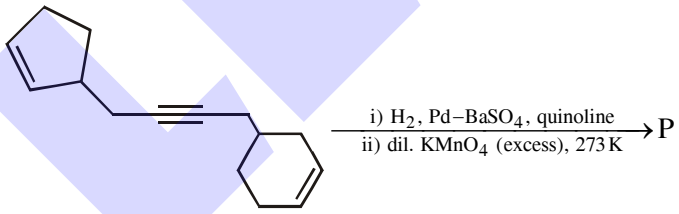

(3) 


(4) 


**OX0047**

**11.** Total number of hydroxyl groups present in a molecule of the major product P is ___

**[JEE-Advance 2019]**



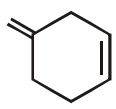

**OX0048**

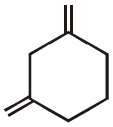
**12.** An unsaturated hydrocarbon X absorbs two hydrogen molecules on catalytic hydrogenattion, and also gives following reaction : **[JEE-MAIN-On-line-(Jan)-2020]**

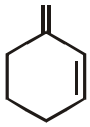
$$X \xrightarrow[Zn/H_2O]{O_3} A \xrightarrow{[Ag(NH_3)_2]^+}$$

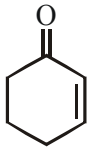
B(3 - oxo - hexanedicarboxylic acid)

X will be :-

(1) 

(2) 

(3) 

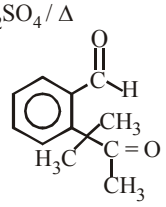
(4) 

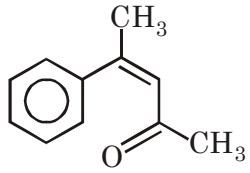

**OX0049**

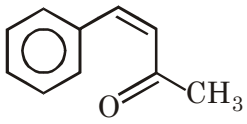
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word Files\Hydr - Carb - Oxi - Red - Ald & Ket - (OC)\Eng\05_Oxidation\01_Theory+Ex.p65

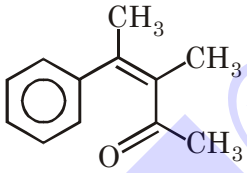
**13.** Identify (A) in the following reaction sequence : **[JEE-MAIN-On-line-(Jan)-2020]**

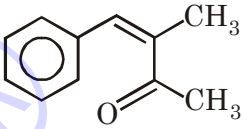
(A) Gives positive iodoform test $\xrightarrow[\text{(ii) } H^+, H_2O \text{ (iii) Conc. } H_2SO_4/\Delta]{\text{(i) } CH_3MgBr}$ (B) $\xrightarrow{O_3/Zn,\ H_2O}$



(1) 

(2) 

(3) 

(4) 

**OX0050**

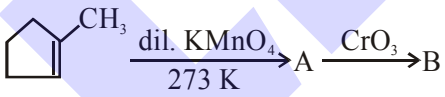
**14.** Which of the following reagent is used for the following reaction ? **[JEE-MAIN--2021]**

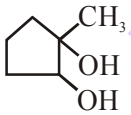
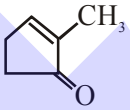
$$CH_3CH_2CH_3 \xrightarrow{?} CH_3CH_2CHO$$

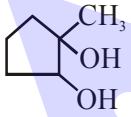
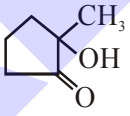
(1) Manganese acetate
(2) Copper at high temperature and pressure
(3) Molybdenum oxide
(4) Potassium permanganate

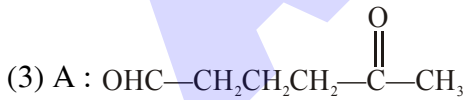
**OX0051**

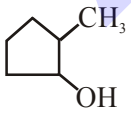
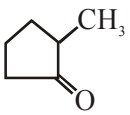
**15.** Identify products A and B : **[JEE-MAIN-2021]**



(1) A :  B : 

(2) A :  B : 

(3) A :  B : 

(4) A :  (4) 

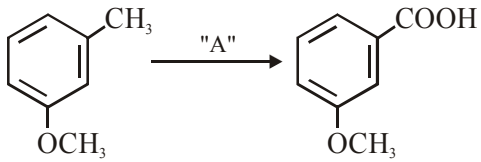
**OX0052**

**16.** Which one of the following reactions will not form acetaldehyde? **[JEE-MAIN--2021]**

(1) $CH_3CH_2OH \xrightarrow[573K]{Cu}$
(2) $CH_3CN \xrightarrow[(ii)\,H_2O]{(i)\,DIBAL-H}$
(3) $CH_2{=}CH_2 + O_2 \xrightarrow[H_2O]{Pd(II)/Cu(II)}$
(4) $CH_3CH_2OH \xrightarrow{CrO_3-H_2SO_4}$

**OX0053**

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hdb-Alc & Et - Hydr - Carb - Oxi - Red - Ald & Ket-(OC)\Eng\05_Oxidation\01_Theory+Ex.p65"

**17.**  **[JEE-MAIN--2021]**

In the above reaction, the reagent "A" is :

(1) $NaBH_4$, $H_3O^+$ (2) $LiAlH_4$

(3) Alkaline $KMnO_4$, $H^+$ (4) HCl, Zn-Hg

**OX0054**

**18.** Given below are two statements : **[JEE-MAIN--2021]**

**Statement-I :** 2-methylbutane on oxidation with $KMnO_4$ gives 2-methylbutan-2-ol.

**Statement-II :** n-alkanes can be easily oxidised to corresponding alcohols with $KMnO_4$.

Choose the correct option :

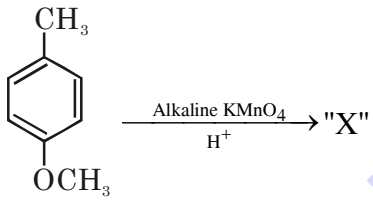
(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

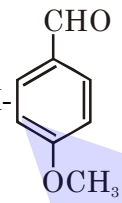
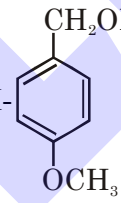
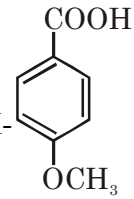
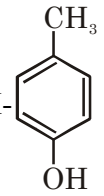
(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct

**OX0055**

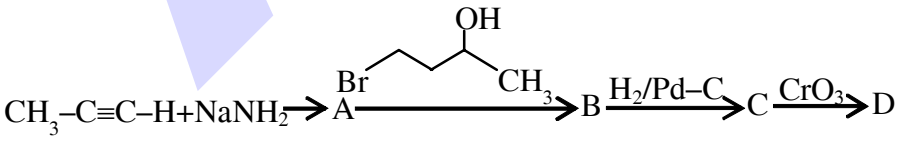
**19.**  **[JEE-MAIN--2021]**

Considering the above chemical reaction, identify the product "X" :

(1) X-  (2) X-  (3) X-  (4) X- 

**OX0056**

**20.** In the following sequence of reactions, the final product D is : **[JEE-MAIN--2021]**



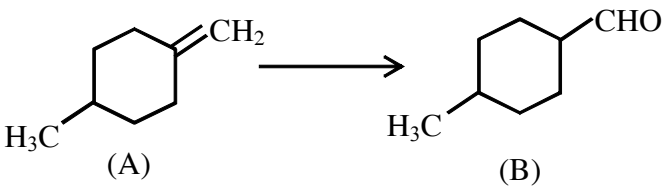
(1) $H_3C–CH_2–CH_2–CH_2–CH_2–\overset{O}{\overset{\|}{C}}–H$ (2) $CH_3$-CH=CH-$CH_2$-$CH_2$-$CH_2$-COOH

(3) $H_3C$-CH=CH-CH(OH)-$CH_2$-$CH_2$-$CH_3$ (4) $CH_3–CH_2–CH_2–CH_2–CH_2–\overset{O}{\overset{\|}{C}}–CH_3$

**OX0057**

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#\Halo-Alc.&Et.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket-(OC)\Eng\05_Oxidation\01_Theory+Ex_p65"

**21.** Which of the following reagents/ reactions will convert 'A' to 'B'? **[JEE-MAIN--2022]**




(1) PCC oxidation

(2) Ozonolysis

(3) $BH_3, H_2O_2$ / $^-OH$ followed by PCC oxidation

(4) HBr, hydrolysis followed by oxidation by $K_2Cr_2O_7$ .

**OX0058**

**22.** 'A' and 'B' respectively are: **[JEE-MAIN--2022]**

**A** $\xrightarrow[(2)\,Zn-H_2O]{(1)O_3}$ Ethane-1,2-dicarbaldehyde + Glyoxal/Oxaldehyde

**B** $\xrightarrow[(2)\,Zn-H_2O]{(1)O_3}$ 5-oxohexanal

(1) 1-methylcyclohex-1, 3-diene & cyclopentene

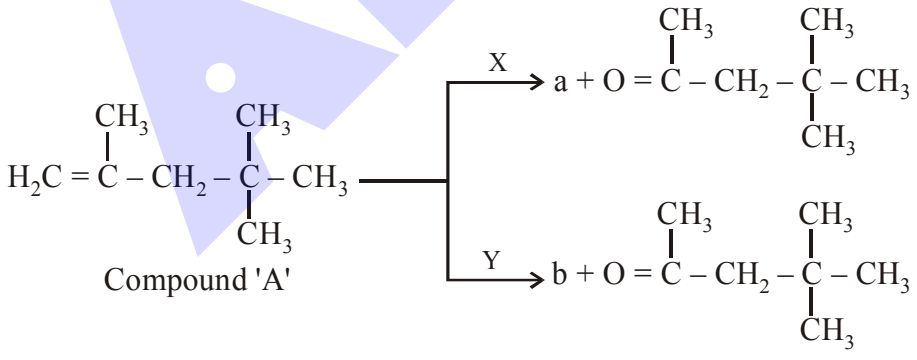
(2) Cyclohex-1, 3-diene & cyclopentene

(3) 1-methylcyclohex-1,4-diene & 1-methylcyclopent-1-ene

(4) Cyclohex-1,3-diene & 1-methylcyclopent-1-ene

**OX0059**

**23.** A compound 'A' on reaction with 'X' and 'Y produces the same major product but different by product 'a' and 'b'. Oxidation of 'a' gives a substance produced by ants. **[JEE-MAIN--2022]**




'X' and 'Y' respectively are :

(1) $KMnO_4/H^+$ and dil. $KMnO_4$, 273 K

(2) $KMnO_4$,(dilute), 273 K and $KMnO_4/H^+$

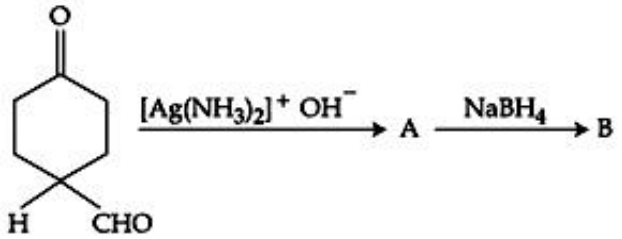
(3) $KMnO_4/H^+$ and $O_3$, $H_2O/Zn$

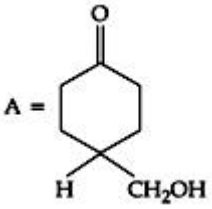
(4) $O_3$, $H_2O/Zn$ and $KMnO_4/H^+$

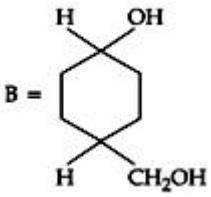
**OX0060**

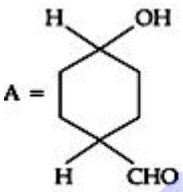
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hdb-Alc & Et- Hydr.-Carb.-Oxi.-Red.-All& Ben-(OC)\Eng\05_Oxidation\01_Theory+Ex.p65"

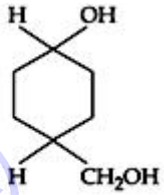
**24.** The products formed in the following reaction, A and B are **[JEE-MAIN--2022]**

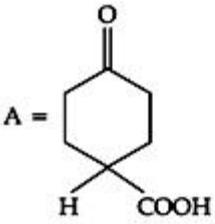


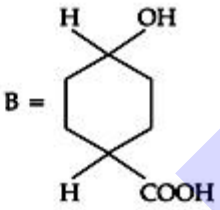

(1) A = 

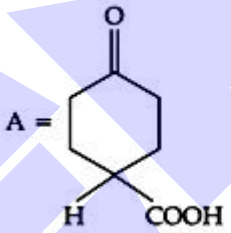
, B = 

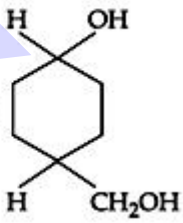

(2) A = 

, B = 


(3) A = 

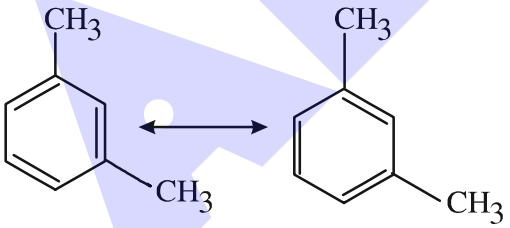
, B = 

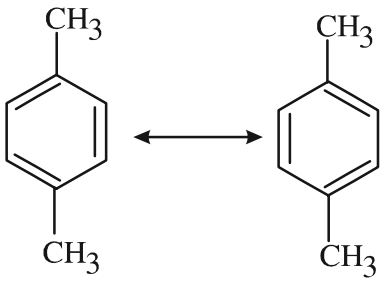

(4) A = 

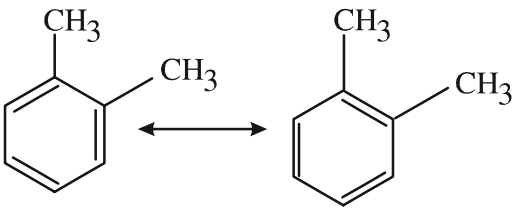
, B = 

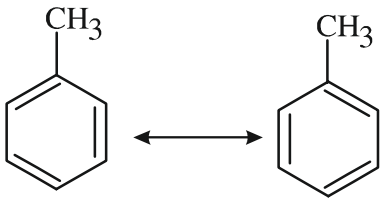

**OX0061**

**25.** Which among the following pairs of the structures will give different products on ozonolysis? (Consider the double bonds in the structures are rigid and not delocalized.) **[JEE-MAIN--2022]**

(1) 


(2) 


(3) 


(4) 


**OX0062**

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Alc.&Et.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket-(OC)\Eng\05_Oxidation\01_Theory+Ex.p65"

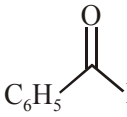
**26.** Hex-4-ene-2-ol on treatment with PCC gives 'A'. 'A' on reaction with sodium hypoiodite gives 'B', which on further heating with soda lime gives 'C'. The compound 'C' is

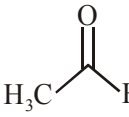
**[ JEE MAINS 2022]**

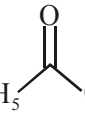
(1) 2- pentene (2) proponaldehyde (3) 2-butene (4) 4-methylpent-2-ene

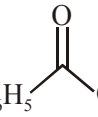
**OX0063**

**27.** Which of the following conditions or reaction sequence will NOT give acetophenone as the major product ? **[ JEE MAINS 2022]**

(1) (a)  (b) $Na_2Cr_2O_7$, $H^+$

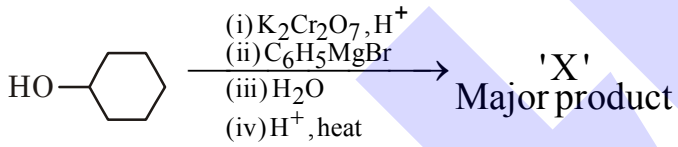
(2) (a)  (b) PCC, DCM

(3) 

(4) 

**OX0064**

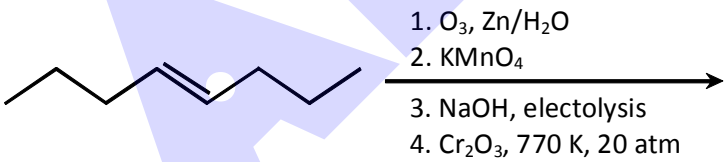
**28.** In the given reaction **[JEE-MAIN--2022]**




the number of $sp^2$ hybridised carbon (s) in compound 'X' is_____.

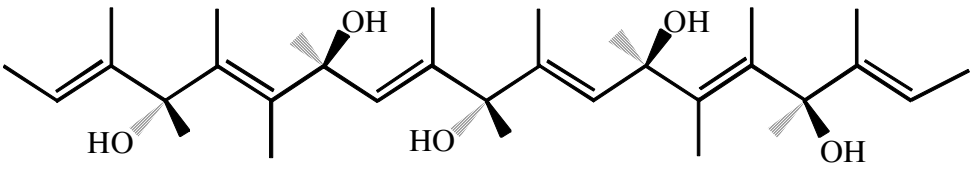
**OX0065**

**29.** The number of -$CH_2$- (methylene) groups in the product formed from the following reaction sequence is _____. **[JEE ADVANCE-2022]**




**OX0066**

**30.** The total number of chiral molecules formed from one molecule of P on complete ozonolysis ($O_3$, $Zn/H_2O$) is _____. **[JEE ADVANCE-2022]**




**P**

**OX0067**

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hdb-Alc & Et- Hydr - Carb -Oxi - Red - Ald & Ket-(OC)\Eng\05_Oxidation\01_Theory+Ex.p65"

# ANSWER-KEY

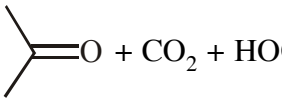
## EXERCISE # O-1

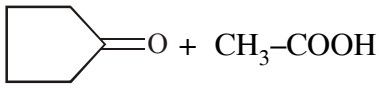
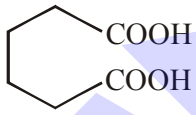
| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | D | B | A | B | D | C | A,C,D | B,D | A,B,C | A,B |
| Que. | 11 | 12 | 13 | 14 | 15 | | | | | |
| Ans. | C | B | C | D | C | | | | | |

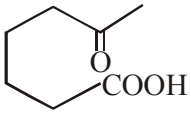
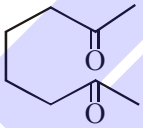
## EXERCISE # O-2

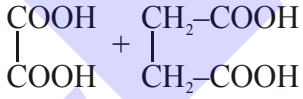
**1.** **(A,C,D)** **2.** **(C)**

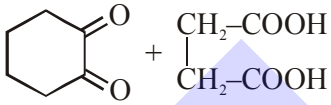
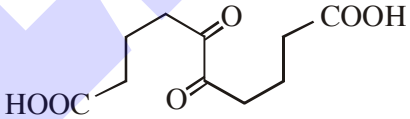
**3.** (i) $CO_2$ (ii) $CH_3–COOH + CO_2$

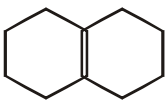
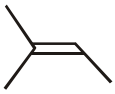
(iii)  $+ CO_2 + HOOC–C–C–C–COOH$

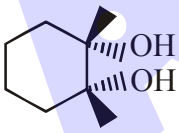
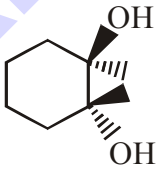
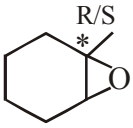
(iv)  $+ CH_3–COOH$ (v) 

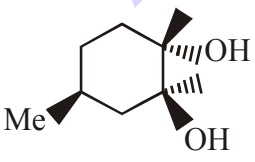
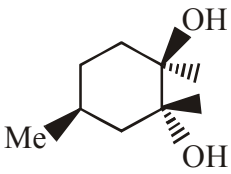
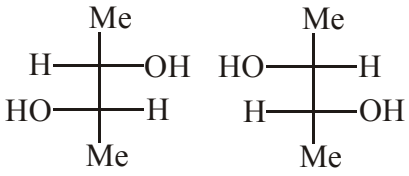
(vi)  (vii) 

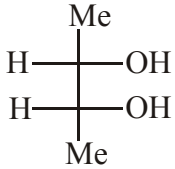
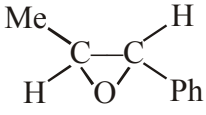
(viii) $2CH_3–COOH$ (ix) 

(x)  (xi) 

**4.** (i) $Me–CH=CH–CH_3$ (ii)  (iii)  (iv) 

**5.** (i) (A)  (B)  (ii) 

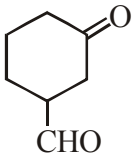
(iii)   (iv) 

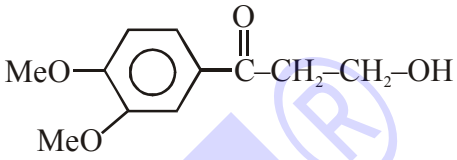
(v)  (vi) 

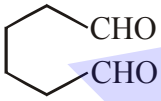
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#\Hdo-Alc & Eth - Hydr - Carb - Oxi - Red - Ald & Ket (OC)\Eng\05_Oxidation\01_Theory+Ex_p65"

**6.** (i) $CH_3–CH_2–COO^-$ , $CH_3–CH_2–COOH$ (ii) $CH_3–\overset{O}{\overset{\|}{C}}–CH_2–CH_3$

**7.** (i) $CH_2=CH–(CH_2)_3–CHO$ (ii) Ph–CH=CH–CHO

(iii) (A) $H_3C–\overset{O}{\overset{\|}{C}}–CH_2–CH_2–CHO$ (iv) 

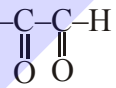
(v) $H_2C=CH–CHO$ (vi) 

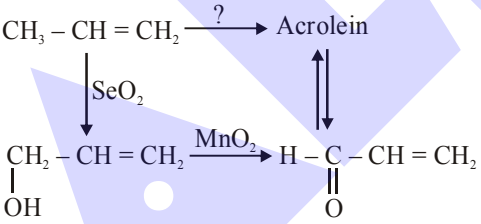
**8.** (i) Me–CHO + HCHO (ii) 

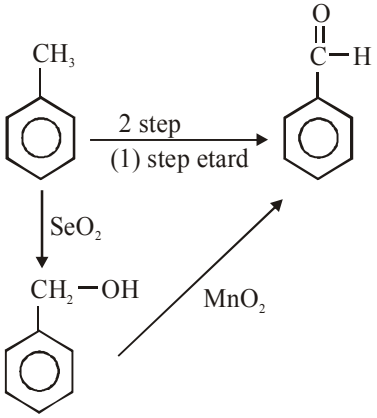
(iii) $HO–CH_2–CH_2–CHO + HCHO$ (iv) $HCHO + HCOOH + CH_3–CHO$

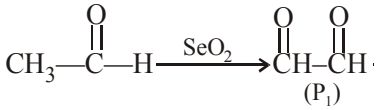
(v) 2HCHO + 2HCOOH (vi) Me–COOH + Me–CHO

(vii) 2Me–COOH

**9.** (a)  (b) 

(c) 

(d) 

(e) 

**10.** **(a)** F and B test

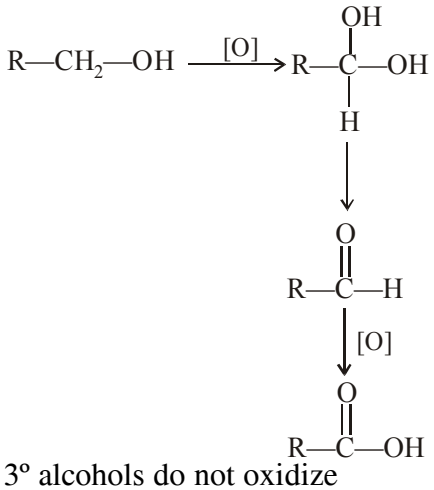
**(b)** Iodoform test

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## EXERCISE# (MAIN & ADVANCE)

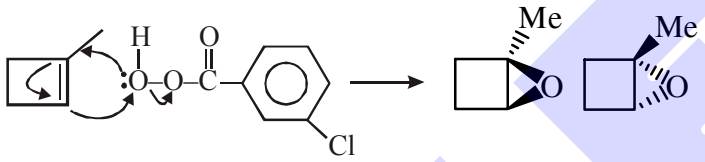
**1.** **(A)** **2.** **(D)**

**3.** **(A)**



3° alcohols do not oxidize
they undergo dehydration

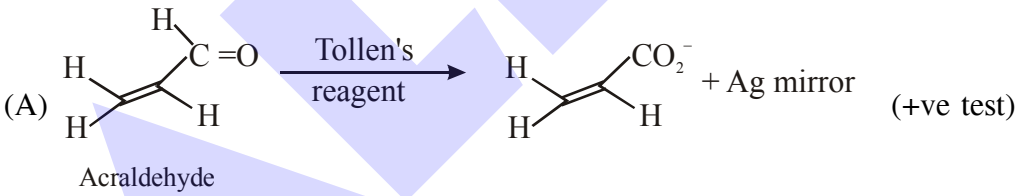
**4.** **(C)**



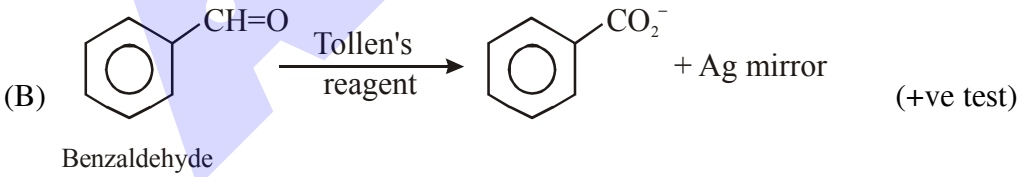
**5.** **(A)** **6.** **(B)**

**7.** **(A,B,C)**

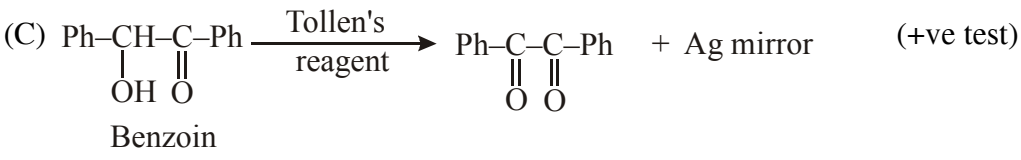
**Sol.** Tollens's test is given by compounds having aldehyde group. Also α-hydroxy carbonyl gives positive tollen's test.



(A) Acraldehyde — Tollen's reagent → + Ag mirror (+ve test)



(B) Benzaldehyde — Tollen's reagent → + Ag mirror (+ve test)



(C) Benzoin — Tollen's reagent → + Ag mirror (+ve test)



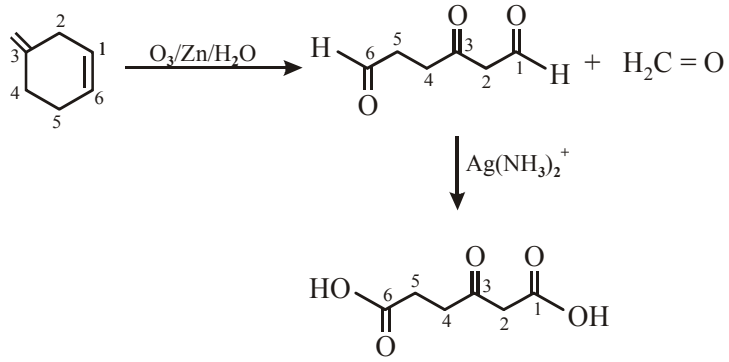
(D) Tollen's reagent → No reaction (–ve test)

**8.** **(D)** **9.** **(2)** **10.** **(4)** **11.** **(6.00)**

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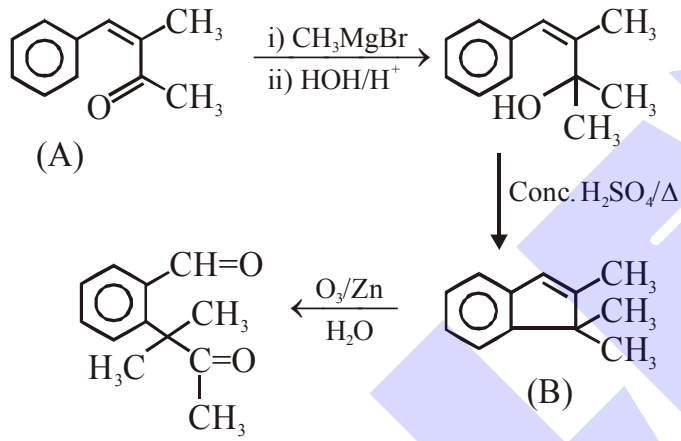
**12. (1)**

**Sol.**




**13. (4)**

**Sol.**




**14. (3)** **15. (2)** **16. (4)** **17. (3)**

**18. (3)** **19. (3)** **20. (4)** **21. (3)**

**22. (2)** **23. (4)** **24. (3)** **25. (3)**

**26. (3)** **27. (3)** **28. (8)** **29. (0)**

**30. (2)**

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